Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years building **TypeScript**, **React**, and **Node.js** applications and production **LLM** workflows with retrieval and multi-model orchestration, plus hands-on observability with Kibana, Sentry, and Azure Monitor.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Web Technologies, Database Systems, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Data Retrieval, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built fault-tolerant **LLM orchestration** with FastAPI background workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs at 3x throughput without cascading failures under concurrent load.
- Architected a multi-tenant platform on Azure SQL and Blob Storage with row-level security for 500+ users, cutting dashboard latency from 1.2s to 350ms and running 5K+ monthly jobs at 99%+ **reliability**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built multi-step **LLM workflows** chaining rubric generation, OCR, grading, and feedback across GPT-4o, Claude, and Qwen-VL behind a React interface, serving 10K+ daily requests at 90%+ consistency.
- Designed **model routing** with timeout fallback and per-call token and cost logging to keep the non-deterministic pipeline reliable, cutting feedback turnaround 60% for 150+ instructors across 1K+ file batches.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Collaborated with Microsoft and GitHub stakeholders across **distributed teams**, running parallel builds on India and US servers and nightly Jenkins test pipelines across Windows, Linux, and macOS.
- Debugged large-scale API integrations using failure logs and coverage reports, improving response times 30% with caching and pagination, and led an ETL redesign boosting **pipeline efficiency** 68%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight **TypeScript** and Node.js microservices with React front ends and MongoDB, delivering real-time WebSocket updates for resources, billing, and leads while syncing 2M+ records across locations.
- Ran production **observability** with Kibana for logs, Sentry for errors, and Azure Monitor for performance metrics, and shipped Docker builds through Jenkins CI/CD at 95%+ build validation.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built Python Flask APIs behind React TypeScript and React Native clients with Firebase **real-time** status sync, sustaining sub-200ms latency and 40% faster confirmations across 50K+ booking records.
- Automated payment event handling with idempotent **webhooks** for refunds, disputes, and chargebacks, cutting manual work 70%, and built hourly Databricks analytics on peak hours and service demand.

Skills

Languages & Frontend: TypeScript, JavaScript, Python, SQL, Java, React, Next.js, Shadcn, TailwindCSS
Backend: Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, Microservices
LLM Systems: LLM Orchestration, Prompt Chaining, Multi-Model Routing, RAG, Embeddings, FAISS, LLM Evaluation, Claude
Observability & Practices: Kibana, Sentry, Azure Monitor, Logging, Monitoring, Claude Code, Code Reviews, Open Source, Distributed Teams
Data & Cloud: PostgreSQL, MongoDB, Azure, AWS, Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins

Projects

Incident Root-Cause Agent | *TypeScript, Node.js, React, Elasticsearch, Claude, OpenTelemetry*

- Designed a TypeScript investigation agent that queries **Elasticsearch** logs and metrics through tools, flags error-rate shifts against baselines, reasons over evidence in a React timeline, and is scored on injected-incident **evals**.

Big-Brain Live | *LLaMA-2, FAISS, Embeddings, React, Shadcn*

- Built a LLaMA-2 **retrieval** assistant with transformer embeddings and FAISS over 50K+ documents, managing context for summarization and Q&A in a React and Shadcn UI at 95% query accuracy.

Chat2SQL Code | *TypeScript, Java, Docker, PostgreSQL*

- Extended the open-source Chat2DB codebase in Java and **TypeScript** into an AI-assisted SQL workspace for 10+ databases, generating schema-aware queries at 82% accuracy with Docker-based local deployment.